

This type of communication via GEO satellites proved very profitable. India started deploying regional GEO system INSAT-1 in 1982 for telecommunication, television distribution and meteorology.

There was a rapid development of GEO satellite systems for international, regional and domestic telephone traffic and TV distribution in the 1970s and 1980s. In the United States, the expansion of fibre-optic links with very high capacity and low delay caused virtually all telephone traffic to move to terrestrial circuits by 1985. However, demand for satellite systems steadily grew through this period and the available spectrum in C band was quickly occupied, leading to expansion into Ku band.

In the US, most of the expansion after 1985 was in the areas of television and business communication (VSAT) networks. In India, however, 95 per cent of the telephone traffic moved to terrestrial circuits during 1990s with 5 per cent still on satellite to meet the requirements of islands and remote areas. Today, in India, the bulk of traffic through communication satellites is for television, VSATs, education and telemedicine.

Direct broadcast satellite

The world's first experimental direct broadcast satellite, ATS-6 (Applications Technology Satellite-6), was launched in May 1974 and it was also the world's first educational satellite. Satellite Instructional Television Experiment (SITE) was conducted using this satellite jointly by NASA and ISRO.

The main mission of ATS-6 was to demonstrate the feasibility of direct-to-home (DTH) television broadcasting. To this end, in addition to the high-gain antenna, the spacecraft payload was able to receive in any of the VHF-, C-, S- and L-bands, and to transmit in S-band (2GHz) through a 20W solidstate trans-



ATS-6 (Applications Technology Satellite-6)

mitter, in L-band (1650MHz) through a 40W transmitter, in UHF (860MHz) through an 80W transmitter (which was used for the SITE experiment in India) and in C-band (4GHz) with a TWTA-based transmitter of 20W. The antenna produced in UHF two spots on earth of 400,000 sq.km each, in which the TV broadcast could be received with 3m diameter antennae.

This payload was first used over the United States for tele-education and telemedicine experiments. The spacecraft was then moved over the geo-stationary arc from 94°W to 35°E, in collaboration with the ISRO, who had deployed television equipment in more than 2500 villages, in six clusters in central India. A tele-education programme was run for one year, during 1975-76. During the experiment, a television receiving equipment was offered by the Indian government to Arthur C. Clarke, who was living in Sri Lanka. This experiment was highly successful, and encouraged ISRO to start building an operational programme, with the Indian spacecraft INSAT 1B (launched 1983). Under SITE

programme, TV broadcasts for school children, school teachers and rural audience on agriculture, health, hygiene, nutrition, etc were made. About 40,000 teachers were trained during SITE and substantial knowledge gains accrued on health and hygiene among the rural population.

The basic definition of direct broadcast satellite (DBS) based on ITU's specifications is that the geostationary satellite receives signals transmitted from the earth and re-transmits them such that inexpensive antennae installed at individual residences spread over a large coverage area receive the signal. The receive system for a DBS system consists of a parabolic dish, a down-converter and necessary equipment for channel selection and decoding.

Due to new allocation of frequency in the Ku and Ka bands, DBS or DTH services became more economically and technically viable as high-power satellites can broadcast in these frequencies to small-sized earth receiving dishes. The best success story of this particular type of satellite service can be found in Europe and Japan. In Europe, this facilitated economic means of sharing television programmes among various nations covered by the footprint of a single satellite, as the cable TV development was slow due to lack of privatisation in the national television system. The Japanese utilised this service to overcome poor TV reception owing to mountainous terrain throughout the island nation.

In India, EDUSAT provides this service to meet the needs of the education sector and reach remote islands and difficult-to-reach terrain. For domestic TV reception, INSAT provides DTH service through the use of high-power Ku-band transponders. One problem with this service is the rain attenuation encountered in Ku-band.